

Tanmoy Sanyal

Scientist, Protein Design & Structural Biology

San Francisco Bay Area, CA

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Computational scientist with over a decade of experience developing physics-based and machine learning methods for protein design and structural biology, with a track record spanning academic research, biotech, and large pharma. Specialized in MLOps, multispecific molecule design, and integrative structural modeling. Demonstrated ability to translate rigorous scientific methods into scalable, cloud-native production systems.

Education

University of California, Santa Barbara

Ph.D., Chemical Engineering — Graduate Emphasis in Computational Science & Engineering (HPC) 2013–2018

Indian Institute of Technology, Kharagpur

Integrated M.Tech & B.Tech (Hons.), Chemical Engineering

2008–2013

Experience

Amgen Inc. — South San Francisco, CA

Principal Protein Design Scientist

Apr 2024–Present

Senior Protein Design Scientist

Oct 2023–Mar 2024

Protein Design Scientist

Jan 2022–Oct 2023

- **Multispecifics design platform:** Developed cloud-native, highly parallelized ODE solvers for pharmacokinetic modeling of bispecific molecules; deployed via ML regression models achieving a **50× speedup**.
- **Statistical mechanics of multispecifics:** Translated statistical mechanical models of superselectivity (originally developed for multivalent ligands) to multispecific molecules; currently undergoing experimental validation and benchmarking on bispecific candidates.
- **MLOps architecture:** Built Amgen's first end-to-end, cloud-native MLOps pipeline for supervised biophysical property prediction from protein sequences, now used across **8+ antibody design projects**. Mentored junior data scientists in thermostability prediction workflows.
- **In-silico display library design:** Developed linear-programming algorithms to optimize degenerate codon libraries using protein language models. Set an internal record for fastest request-to-delivery (~3 days) for a 1M+ diverse antibody library; applied to engineer cross-reactivity across 2 therapeutic areas.
- **Biologics design:** Designed mini-binders for a rare disease target using Rosetta-ddG and the AlphaFold, RFDiffusion, and ProteinMPNN workflow. Developed an active-learning pipeline (Gaussian process regression) yielding a **20×** affinity improvement over 3 design-and-screen iterations.
- **AlphaFold cloud deployment:** Collaborated with AWS to re-tool open-source AlphaFold-2 into a scalable cloud-native version supporting ~100k sequences; now in regular use for variant design.

Novo Nordisk Research Center Seattle — Seattle, WA

Protein Design Scientist

2022–2023

- **Peptide modifications for half-life extension:** Developed a computational method for systematically scanning PEGylation sites on peptides for half-life protraction. Published and open-sourced the method with a benchmark on GLP-1 analogs relevant to anti-obesity therapeutics.

University of California, San Francisco — Sali Lab

Postdoctoral Scholar

Jan 2019–Jan 2022

- **Nanobody biophysics & vaccine design:** Developed a protein–protein docking score for nanobody interactions with the SARS-CoV-2 Spike protein using chemical crosslinking and escape mutation data; identified vulnerable and resilient epitopes across variants of concern. Also performed MD simulation studies elucidating the effects of framework mutations on distal CDR3 residues.

- **Protein structure modeling from crosslinks:** Developed graph-sampling algorithms to extract domain boundaries from chemical crosslinking data; published the first coarse-grained structure of the full SMC5/6 protein complex.
- **Whole-cell digital twin:** Developed Bayesian networks combining pharmacokinetic, Brownian dynamics, and enzyme-pathway models of the pancreatic beta-cell into a proof-of-concept multimodal digital twin. Served as software lead in the Pancreatic Beta-Cell Consortium (UCSF/USC).

University of California, Santa Barbara — Shell Lab

Ph.D. Researcher

2013–2018

- **Coarse-graining for phase separation:** Introduced the local density potential for structurally and thermodynamically accurate coarse-grained MD simulations of liquid–liquid phase separation.
- **Protein folding via variational inference:** Developed backbone models for template-free folding of 200+ residue protein domains using MD simulations, and for studying self-assembly in amyloidogenic peptides.
- **HPC cluster administration:** Set up and administered a 144-core Linux (Rocks 6.2) cluster; wrote utilities for automated resource sharing among lab members.

Computational Skills

Protein Design	Rosetta, RFDiffusion, ProteinMPNN, AlphaFold, ESM2, Amplify (protein language models)
Molecular Modeling	OpenMM, LAMMPS, GROMACS, IMP, Modeller, UCSF ChimeraX, PyMOL
ML / MLOps	PyTorch, PyTorch-Lightning, TensorFlow-Probability, PyMC3, MLflow, AWS Batch
Languages	Python (+Cython, 10k+ lines), C++ (~4k lines), Fortran-90 (~1k lines), Bash

Open-Source Software

- **nnprotscan** — Computational scanning of chemical modifications in peptides.
- **nbspike** — Integrative epitope prediction for nanobody–antigen interactions.
- **IMP-raindrops** — Crosslink-guided domain discovery within the IMP framework.
- **PDB-Dev deposition tools** — Structural deposition utilities for the SMC5/6 complex.
- **LAMMPS local density potential** — Plugin implementing the local density coarse-graining potential.

Selected Publications

- N. Thomas, **T. Sanyal**, P. Greisen Jr. & K. Diebler. Structure-Based Computational Scanning of Chemical Modification Sites in Biologics. *ACS Omega*, 9(34), 36787–36794, 2024.
- F.D. Mast, P.C. Fridy, N.E. Karen, J. Wang, E.Y. Jacobs, J.P. Olivier, **T. Sanyal**, et al. Highly synergistic combinations of nanobodies that target SARS-CoV-2 and are resistant to escape. *eLife*, 10: e73027, 2021.
- B. Raveh, L. Sun, K.L. White, **T. Sanyal**, et al. Bayesian metamodeling of complex biological systems across varying representations. *PNAS*, 118(35) e2104559118, 2021.
- Y. Yu, S. Li, Z. Ser, **T. Sanyal**, et al. Integrative analysis reveals unique structural and functional features of the Smc5/6 complex. *PNAS*, 118(19) e2026844118, 2021.
- **T. Sanyal**, J. Mittal and M. Scott Shell. A hybrid, bottom-up, structurally-accurate, Go-like coarse-grained protein model. *J. Chem. Phys.*, 151(4): 044111, 2019.
- **T. Sanyal** and M. Scott Shell. Coarse-grained models using local-density potentials optimized with the relative entropy: Application to implicit solvation. *J. Chem. Phys.*, 145, 034109, 2016.

Service & Peer Review

Reviewer (2019–Present): *Structure*; *Journal of Physical Chemistry*; *Proteins: Structure, Function and Bioinformatics*; *Review of Scientific Instruments*; *Rapid Reviews COVID-19*; *Life*.